

Recovering reaction kinetics and homeostatic regulation from metabolite dynamics with graph neural networks

Abstract

A metabolic network is a bipartite graph in which metabolites are coupled through reactions, each with its own rate constant and stoichiometry. We pose the *inverse* problem: given only the time series of metabolite concentrations and the bipartite graph structure, recover the components that generated the dynamics — the per-reaction rate constants k_j , the substrate-to-rate kinetic law, and the per-metabolite homeostatic regulation. We replace the unknown functions of the forward model with learnable MLPs that operate on the metabolite–reaction graph and a vector of learnable rate constants, and train the graph neural network (GNN) end-to-end to predict dc/dt . On synthetic mass-action systems with known ground truth we study rate-constant recovery and the identifiability obstacles that arise from scale ambiguity, functional compensation, and homeostasis absorption. We outline the path from synthetic data to real metabolomics — starting from fully parameterised kinetic models from BioModels (yeast glycolysis), constraint-based genome-scale models (yeast-GEM), and high-frequency real-time metabolomics during nutrient transitions. This is a companion methods note to our connectome-dynamics studies, which apply the same sign-/structure-locked GNN inverse-modelling recipe to neural circuits.

1 Introduction

Cellular metabolism is a large, sparse reaction network: hundreds of metabolites are interconverted by hundreds of enzyme-catalysed reactions, each proceeding at a rate set by an intrinsic rate constant and the instantaneous concentrations of its substrates. The *forward* problem — predicting concentration dynamics given the network, the kinetics and the rate constants — is classical biochemistry. The *inverse* problem is harder and is our subject: given measured concentration trajectories and the graph of which metabolites enter which reactions, recover the rate constants and the functional form of the kinetics. This is the metabolic analogue of the connectome-to-dynamics inverse problems we study elsewhere, and it shares their central difficulty: the mapping from parameters to dynamics is nonlinear, high-dimensional, and degenerate — many parameter combinations produce nearly identical trajectories.

2 Metabolic networks as bipartite graphs

A metabolic network is naturally a **bipartite graph** with two node types: metabolites and reactions. Edges connect only nodes of different type, and each edge carries an integer *stoichiometric coefficient*. A single-partite metabolite–metabolite graph cannot represent the system, because a reaction couples *several* substrates and products at once through a shared rate constant: collapsing it to pairwise metabolite edges would lose the information that those metabolites participate in the same reaction.

Stoichiometric matrix. The connectivity is summarised by the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{n \times m}$ (n metabolites, m reactions), where S_{ij} is the signed coefficient with which metabolite i enters reaction j : negative for substrates (consumed), positive for products (produced). $\mathbf{S}$ is sparse (each reaction touches only a handful of metabolites), integer-valued (typically in $\{-2, -1, 0, +1, +2\}$), and for mass-balanced reactions its columns sum to zero.

3 Forward model: reaction kinetics and homeostasis

Mass-action kinetics. Under the law of mass action a reaction proceeds faster when its substrates are more abundant, with each substrate raised to the power of its stoichiometric coefficient. For reaction j with rate constant k_j ,

$$v_j = k_j \prod_{k \in \text{sub}(j)} c_k^{|S_{kj}|}, \quad (1)$$

so a reaction consuming two molecules of a substrate depends quadratically on its concentration, and one consuming a single molecule depends linearly. The product over substrates (*multiplicative* aggregation) encodes the requirement that *all* substrates be present simultaneously, and the resulting nonlinear coupling is what allows autocatalytic cycles to sustain oscillations. A simpler *additive* aggregation $v_j = k_j \sum_k c_k^{|S_{kj}|}$ treats substrates as contributing independently and is a useful surrogate for enzyme-mediated reactions that are not strictly mass-action.

Full dynamics. Combining reaction fluxes through $\mathbf{S}$ with a per-metabolite homeostatic restoring force gives the complete forward model

$$\frac{dc_i}{dt} = \underbrace{-\lambda_i (c_i - c_i^{\text{baseline}})}_{\text{homeostasis}} + \underbrace{\sum_{j=1}^m S_{ij} v_j}_{\text{reaction dynamics}}, \quad (2)$$

with v_j from Eq. (1). The homeostatic term pulls each metabolite toward a baseline level c_i^{baseline} at a metabolite-specific rate λ_i ; crucially it is *small* relative to the reaction fluxes, which is what makes it hard to recover (Sect. 5). Eq. (2) spans a family of regimes used for synthetic data: pure reaction ($\lambda = 0$, additive), homeostatic ($\lambda > 0$, additive, convergent), and oscillatory (autocatalytic 3-cycles, multiplicative, sustained oscillations).

4 Inverse problem: the GNN parameterisation

Given observed trajectories $\{c_i(t)\}$ and the bipartite graph, we recover the generating model by replacing every unknown function in Eqs. (1) and (2) with a learnable component on the graph:

$$\frac{dc_i}{dt} = \underbrace{\text{MLP}_{\text{node}}(c_i, a_i)}_{\approx -\lambda_i (c_i - c_i^{\text{baseline}})} + \sum_{j=1}^m S_{ij} k_j \underbrace{\prod_{k \in \text{sub}(j)} \text{MLP}_{\text{sub}}(c_k, |S_{kj}|)}_{\approx k_j \prod_k c_k^{|S_{kj}|}}. \quad (3)$$

The GNN operates on the metabolite–reaction graph by message passing: substrate states are mapped through MLP_{sub} , aggregated per reaction (multiplicatively for mass-action), scaled by the learnable

rate constant k_j , and distributed back to metabolites through $\mathbf{S}$; the node update MLP_{node} adds the self-regulation term. Training minimises the prediction error on dc/dt end-to-end. The learnable parameters are listed in Tab. 1.

Parameter	Type	Role
$a_i \in \mathbb{R}^d$	embedding vector	per-metabolite identity
k_j	scalar (m of them)	per-reaction rate constant (learned in log-space)
$\text{MLP}_{\text{sub}}(c_k, S_{kj})$	MLP	discovers the kinetic law $c_k^{ S_{kj} }$
$\text{MLP}_{\text{node}}(c_i, a_i)$	MLP	discovers homeostasis $-\lambda_i(c_i - c_i^{\text{baseline}})$

Table 1: Learnable components of the GNN inverse model (Eq. (3)). The stoichiometric matrix $\mathbf{S}$ is either frozen from ground truth (“S given” mode, the focus here) or made learnable (“S learning” mode).

Function discovery. The GNN is not told the kinetic law. MLP_{sub} receives a substrate concentration and its stoichiometric coefficient and must learn the power-law response $c_k^{|S_{kj}|}$ from data alone — in particular it must distinguish linear (c^1) from quadratic (c^2) dependence as the coefficient changes. Likewise MLP_{node} must recover the subtle, small-magnitude homeostatic signal without absorbing dynamics that belong to the reaction terms.

5 Identifiability

The three learnable components interact and can compensate for one another, producing degenerate solutions with high prediction accuracy but poor parameter recovery:

- **Scale ambiguity.** The product $k_j \text{MLP}_{\text{sub}}$ is invariant under $k \rightarrow \alpha k$, $\text{MLP}_{\text{sub}} \rightarrow \text{MLP}_{\text{sub}}/\alpha$. Without anchoring, the MLP absorbs a global scale and shifts every k_j . A centring regulariser pins $\text{mean}(\log k)$ to the known range centre.
- **Functional compensation.** If MLP_{sub} settles on a wrong exponent (*e.g.* $c^{1.5}$ instead of c^2), the rate constants can partly compensate by re-scaling, masking the error.
- **Homeostasis absorption.** MLP_{node} can grow and soak up dynamics that should be explained by reactions, biasing the recovered rate constants.

These are the metabolic counterparts of the sign/scale degeneracies that motivate the structure-locking and regularisation priors in our connectome-dynamics models.

6 Toy model

The toy model is a synthetic rank-50 oscillatory mass-action network (100 metabolites, 256 reactions, $\mathbf{S}$ given, no homeostasis), generated with known rate constants and fit leak-free (the loss reads only the concentration dynamics, never a ground-truth label). Rate-constant recovery R^2 (learned vs. true $\log_{10} k_j$) is 0.75 ± 0.04 across four seeds (Tab. 2), and the two learned functions match ground truth (Fig. 1): MLP_{sub} recovers $c^{|s|}$ and MLP_{node} stays ≈ 0 . The free rollout tracks the oscillations

early but, with no multi-step constraint, the autocatalytic dynamics eventually diverge: this model recovers the parameters but is not a stable simulator.

Does a recurrent curriculum (growing the rollout horizon across epochs, single-step $\rightarrow$ 1000 steps, with co-ramped LR and gradient clipping) close the stability gap? A naive run is a cautionary tale about the metric. The free trajectory no longer diverges and the *pooled* rollout Pearson even rises, but the pooled number merely rewards staying bounded; measured *per metabolite*, fidelity *collapses*, and rate-constant recovery collapses with it (Fig. 2) while MLP_{node} drifts to a spurious nonzero function. The multi-step objective admits a smooth *degenerate* solution — wrong k plus spurious homeostasis reproduce a damped trajectory without the mechanism. The lesson: report *per-metabolite* rollout fidelity, and recognise that the naive curriculum suppresses dynamics rather than buying stability.

Seed	training completed	rate-constant recovery R^2	rollout Pearson
77 (published winner)	100%	0.742	0.74
42	100%	0.705	0.77
79	90%	0.808	0.37
7	90%	0.738	0.72
<i>mean $\pm$ s.d. (4 mature seeds)</i>		0.748 $\pm$ 0.037	0.65 $\pm$ 0.16
123	30% (stopped)	0.722 (rising)	0.65

Table 2: Cross-seed reproduction of the leak-free k_j -recovery working point. Rate-constant recovery R^2 clusters tightly at 0.75 ± 0.04 over four seeds; seed 123 was stopped early at 0.72 and still rising. Rollout Pearson is the global correlation between the learned free rollout and the true trajectory over the convergent window (before divergence); it is far more variable (0.65 ± 0.16) and never stable — every seed recovers the rate constants but none yields a stable simulator.

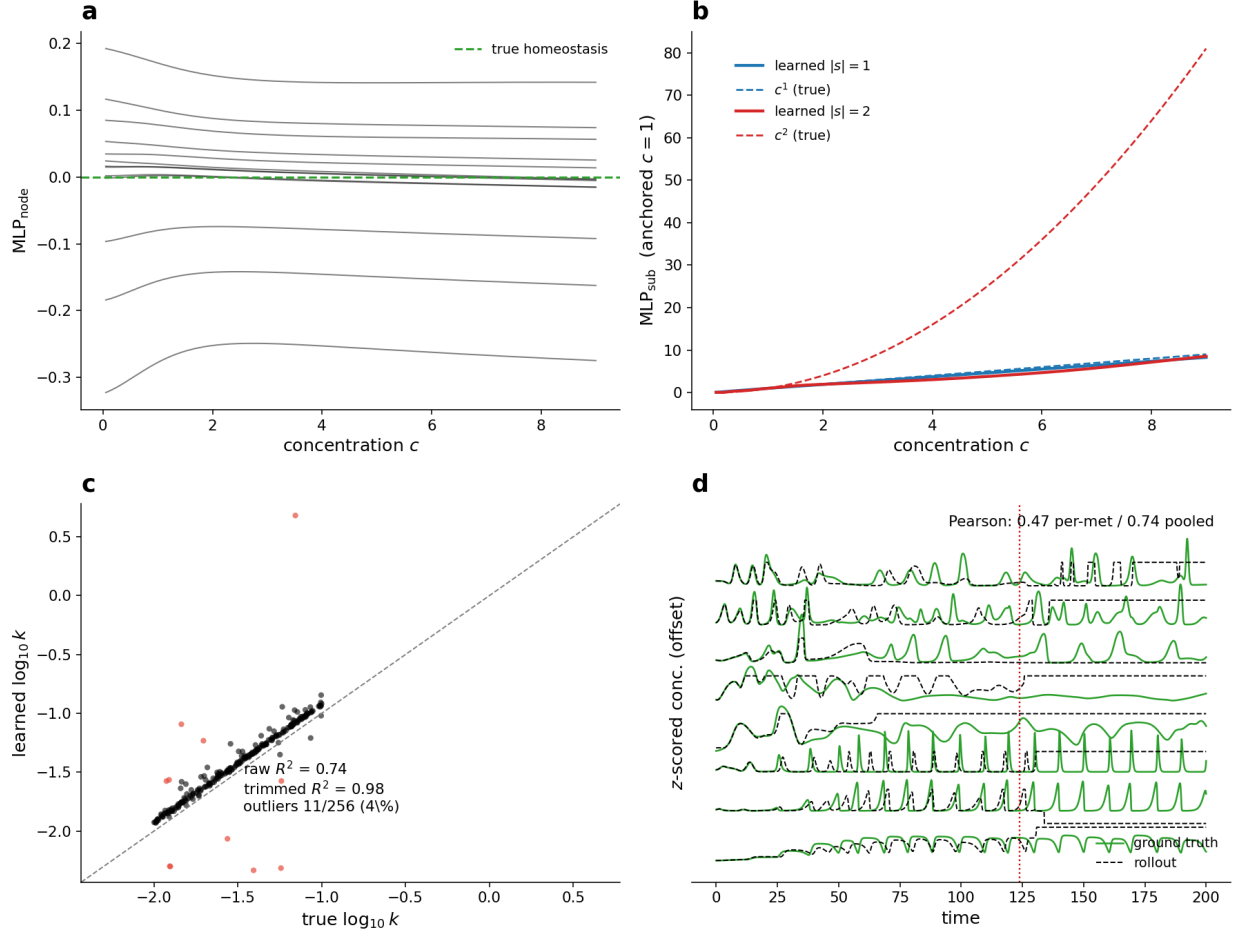


Figure 1: **Toy-model results dashboard (leak-free, seed 77).** (a) MLP_{node} per metabolite (homeostasis; dashed: the true zero of this regime). (b) $MLP_{sub}(c, |s|)$ (solid) vs. the true power law $c^{|s|}$ (dashed) for $|s|=1, 2$, anchored at $c=1$ ($|s|=2$ is out-of-support extrapolation). (c) Rate-constant recovery: learned vs. true $\log_{10} k_j$ for all 256 reactions after the MLP_{sub} scalar correction (dashed: identity); red points exceed the 0.3-dex outlier gate. (d) Free rollout of the eight highest-variance metabolites, z-scored and stacked: ground truth (green) vs. the learned autoregressive rollout (dashed black); red line marks rollout divergence.

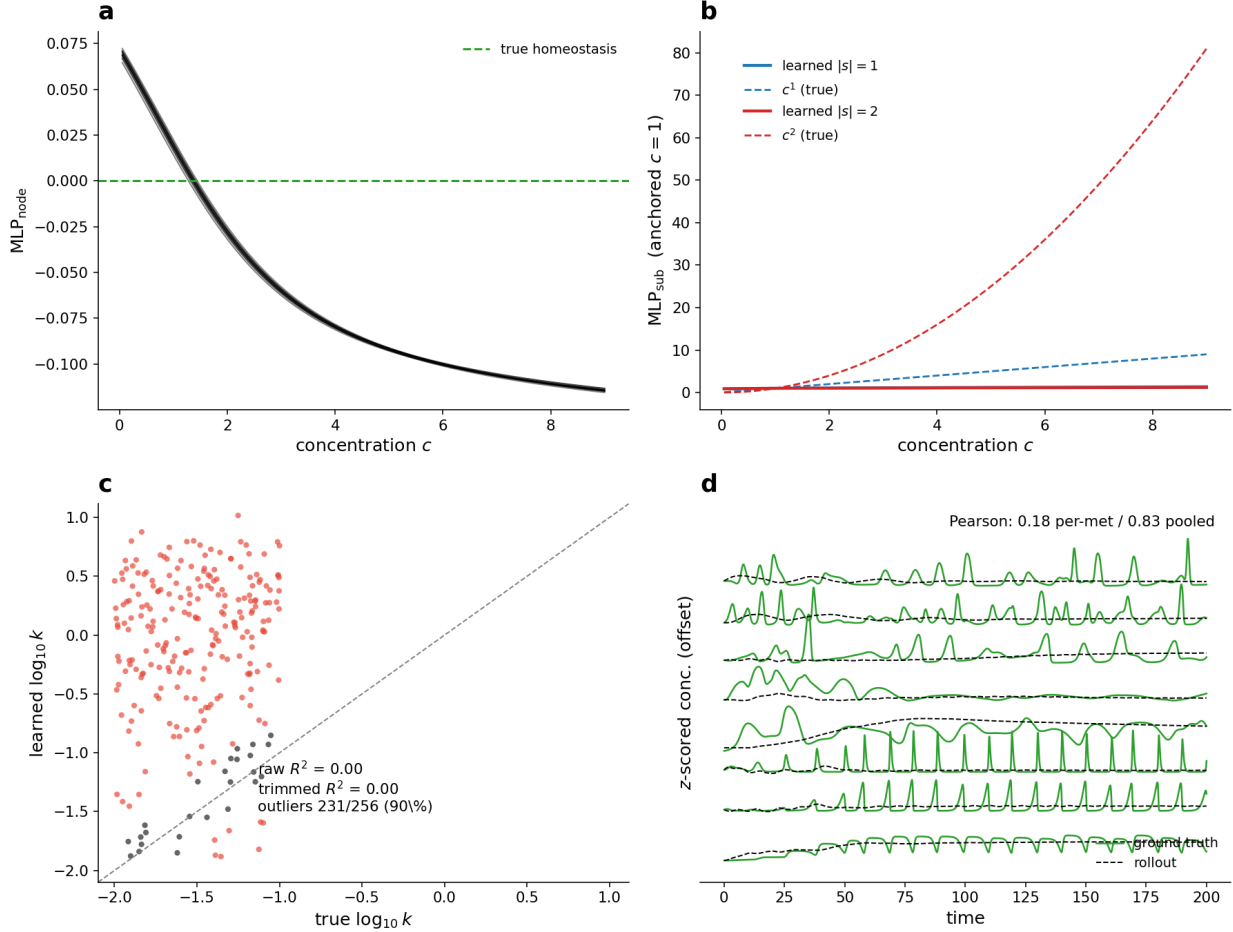


Figure 2: **Naive recurrent curriculum: stable but dynamically dead.** Same dashboard as Fig. 1, for a model trained with the horizon-growing curriculum (single-step $\rightarrow$ 1000 steps). **(d)** The free rollout no longer diverges but the predicted traces are smooth and bounded, no longer reproducing the oscillations; **(c)** rate-constant recovery has collapsed to a structureless cloud; and **(a)** MLP_{node} has drifted to a spurious nonzero homeostasis. **(b)** $|s|=2$ is out-of-support extrapolation, as in Fig. 1.)

A different lever closes the stability gap: supplying a known *external stimulus*. Feeding a time-varying drive to a subset of metabolites at every rollout step — the boundary-input convention of the calcium models — changes the picture. With the drive supplied and ten driven trajectories for training, even *single-step* training gives a rollout that is stable over the full window while preserving rate-constant recovery, and stays bounded on a held-out trajectory with a different drive (Fig. 3). Running the stimulus-driven *curriculum* on top (three horizon schedules to a fixed 12h budget; Fig. 4) does not improve on this: the stimulus prevents the dead-collapse the stimulus-free curriculum suffered, but once training reaches a multi-step horizon the curriculum erodes k -recovery without a rollout gain — the same identifiability tension as Fig. 2, softened by the drive. **Single-step training with the external stimulus remains the best toy configuration.**

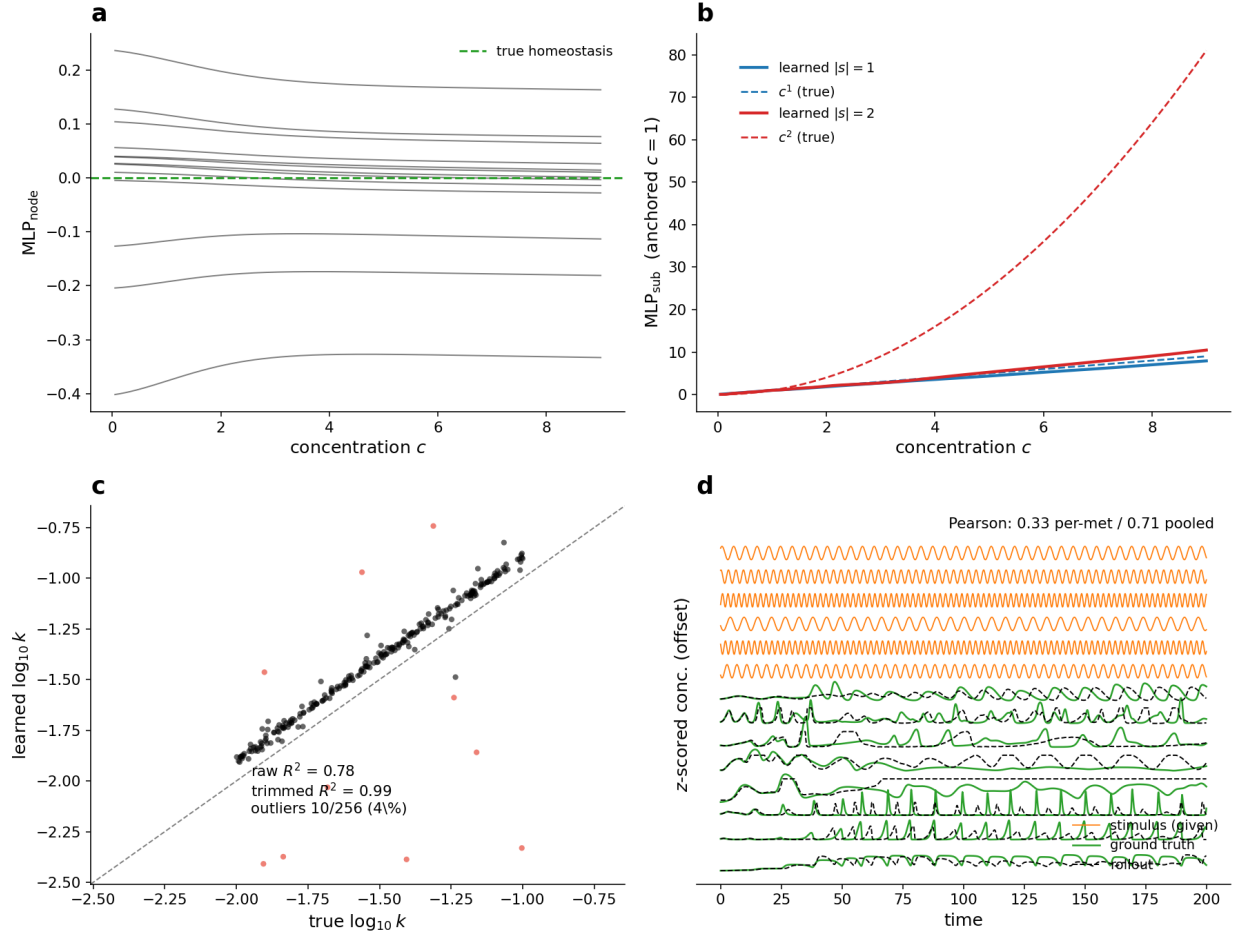


Figure 3: **An external stimulus anchors the rollout.** Same dashboard, for single-step training on ten trajectories driven by a known time-varying stimulus (orange in **d**, the given input). **(d)** The free rollout (black) is now stable over the full window with no divergence (contrast Fig. 1d and Fig. 2d); **(c)** rate-constant recovery is preserved — the stimulus stabilises the rollout without the parameter collapse the curriculum caused.

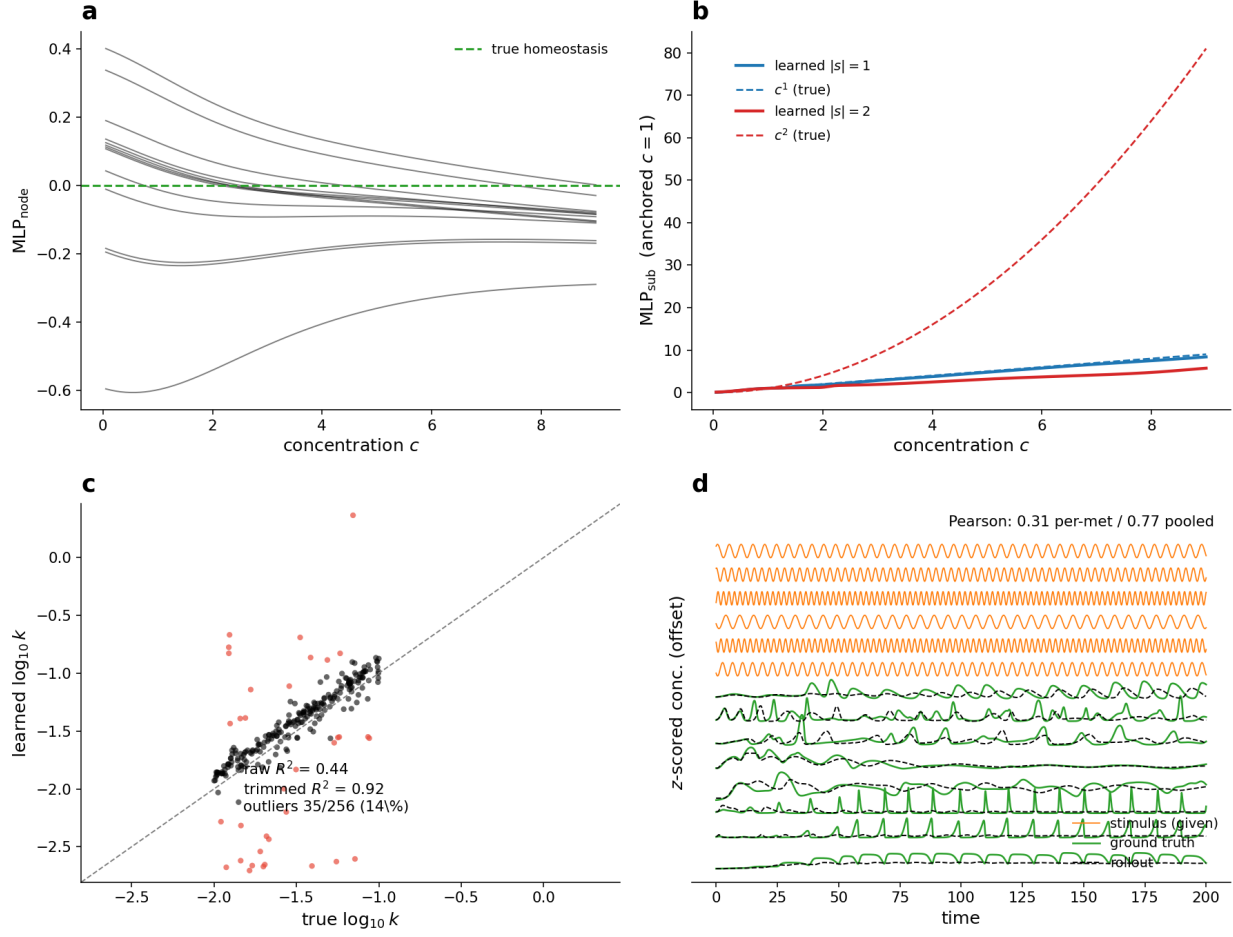


Figure 4: **The recurrent curriculum, even anchored by the stimulus, does not beat single-step training.** Best of three stimulus-driven curriculum schedules (capped cap-100 horizon, 12 h budget). **(d)** The free rollout (black) under the given drive (orange) is stable and tracks the ground truth (green), but **(c)** rate-constant recovery has degraded (red outliers) and **(a)** MLP_{node} drifts to a spurious nonzero function — the multi-step objective trades away k identifiability for no rollout gain.

Intrinsic noise as an identifiability lever — and the right form for a conserved system.

Stochastic forcing *inside* the dynamics can diversify a trajectory and help break a degeneracy, but its *form* matters under a conservation law. Adding noise directly to the concentrations ($dc = f(c)dt + \sigma dW$) injects mass no reaction produced — stoichiometrically unsound. The sound alternative perturbs the reaction *rates*, $k_j(t) = k_j(1 + \sigma \xi_j(t))$ (enzyme-activity fluctuation): mass is conserved through S , trajectories stay physical, and the data diversify (Fig. 5). But here this does not change the outcome: with S given, k is *already* identifiable at $\sigma = 0$, so there is no degeneracy left to break, and the noise is benign at low level then harmful as it corrupts the trajectory. The lever would matter only in a genuinely k -degenerate regime (e.g. S unknown), left to future work.

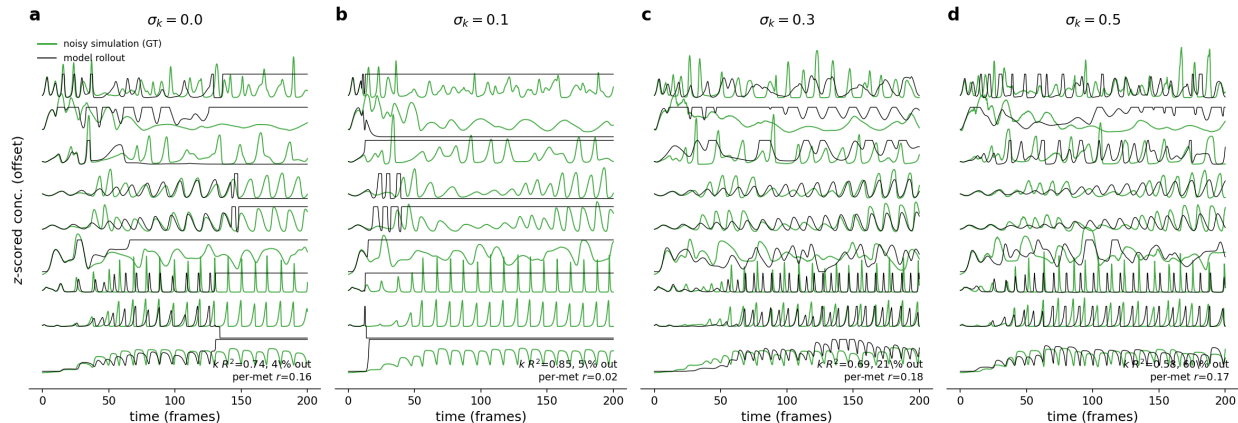


Figure 5: **Stoichiometrically-sound rate-noise sweep (rollout traces)**. Intrinsic noise applied to the reaction *rates*, $k_j(1 + \sigma_k \xi)$ (enzyme-activity fluctuation; mass-conserving), at four levels. Each panel shows the noisy simulated ground truth (green) and the trained model’s deterministic free rollout (black), z-scored and stacked, with k -recovery R^2 and per-metabolite rollout Pearson annotated. As σ_k grows the simulation becomes visibly jagged while the model rollout stays smooth.

7 Toward real metabolomics

7.1 First look: are the four real datasets amenable?

Before training, we run the same cheap four-probe test on *all four* datasets measured by Link et al. (2015) — *E. coli*, *B. subtilis*, *Y. lipolytica*, and a hybrid sample — asking whether each recording carries the structure the GNN needs (Tab. 3, Fig. 6).

What the three probes compute, and how to read them.

- **SVD low-rank** (rank_{99} , the number of modes for 99% of the z-scored variance): the effective dimensionality of the dynamics. A small rank_{99}/N means coordinated, low-dimensional motion a compact network model can fit; $\approx N$ means no parsimonious mechanism exists.
- **Per-ion lag-1 autocorrelation** ρ_1 (median over ions): temporal smoothness. $\rho_1 \rightarrow 1$ is a smooth trace whose derivative dc/dt carries signal; $\rho_1 \rightarrow 0$ is white-noise-like, so dc/dt is noise. This is the decisive test, since the GNN learns $dc/dt = f(c)$.
- **KEGG coverage**: the fraction of ions mapping (by m/z annotation) onto a reference network (iJO1366) — whether a stoichiometry $\mathbf{S}$ can be built for the measured species at all.

Organism	ions $\times$ t	rank ₉₉	autocorr med.	in iJO1366	dc/dt
<i>E. coli</i>	247 $\times$ 119	102 (41%)	0.27	190 (77%)	noisy
<i>B. subtilis</i>	239 $\times$ 119	98 (41%)	0.41	160 (67%)	noisy
<i>Y. lipolytica</i>	164 $\times$ 59	51 (31%)	0.62	103 (63%)	smooth
Hybrid	338 $\times$ 60	55 (16%)	0.24	155 (46%)	noisy

Table 3: Amenability of the four measured datasets. rank₉₉ is the number (and fraction) of singular components for 99% variance; autocorr is the median per-ion lag-1 autocorrelation; coverage is the count mapped onto iJO1366. All four are low-rank and reasonably network-mappable; only *Y. lipolytica* has smooth (dc/dt -learnable) dynamics.

Does it generalise? Two signals hold across all four datasets — the dynamics are low-rank and a large share of metabolites are core/conserved — but the third, temporal smoothness of dc/dt , does *not*: *E. coli*, *B. subtilis* and the hybrid are noise-dominated, whereas ***Y. lipolytica* is smooth**. The actionable consequence: the most promising real-data target is not *E. coli* (best coverage but noisiest) but *Y. lipolytica*, restricted to its smooth, network-mapped ions and judged by held-out prediction.

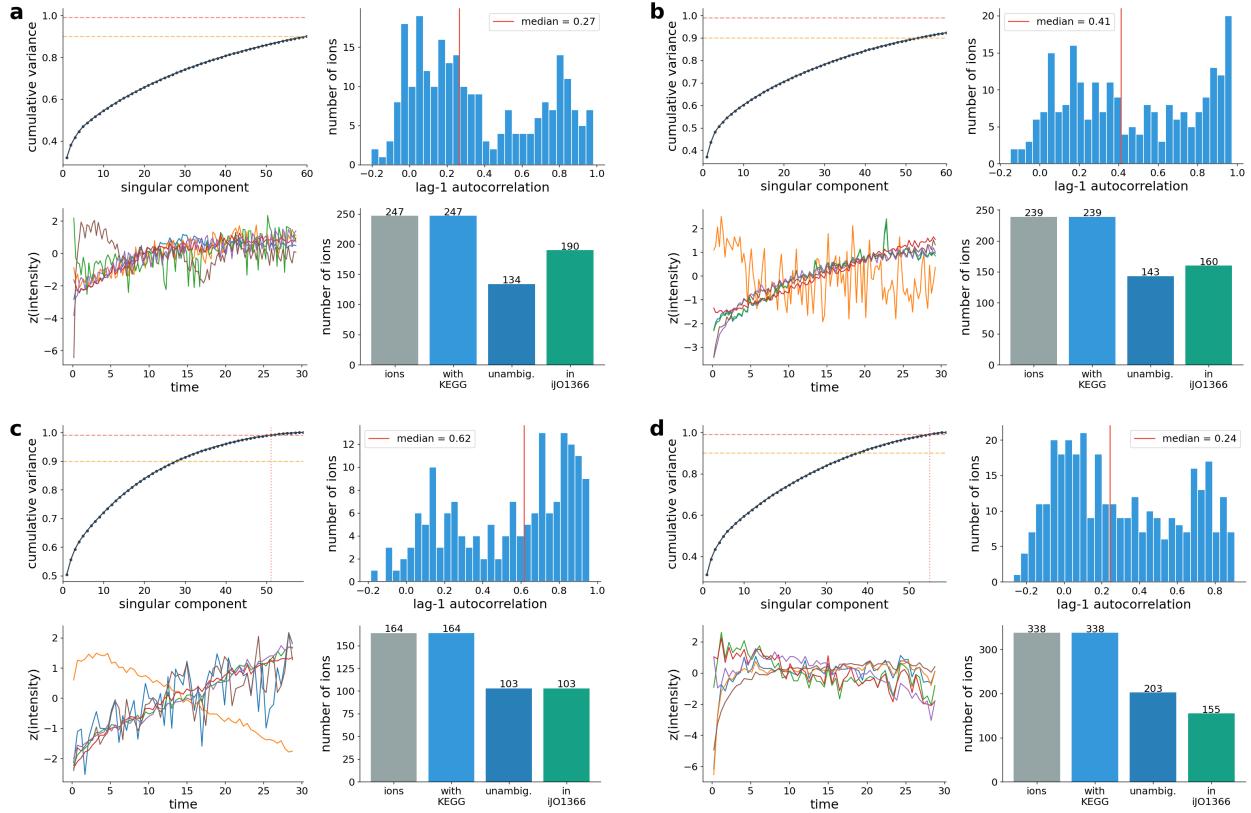


Figure 6: **Identical amenability probes across the four measured datasets:** (a) *E. coli*, (b) *B. subtilis*, (c) *Y. lipolytica*, (d) hybrid. Within each panel, top-left: SVD cumulative variance; top-right: per-ion lag-1 autocorrelation; bottom-left: six highest-variance traces (z -scored); bottom-right: KEGG coverage against iJO1366.

The framework is validated on synthetic data with known ground truth; the goal is real metabolomic time series. Three vendored datasets stage that transition from fully parameterised kinetics, through

genome-scale topology, to a real recording. None is solved yet; this section describes what each provides and what stands between it and a result.

Rung 1 — yeast glycolysis kinetic model (BioModels the published yeast glycolysis kinetic model). A small, fully parameterised kinetic model of glycolysis in *S. cerevisiae* (Smallbone et al., 2013), imported from SBML: 20 internal metabolites (+9 fixed boundary species), 30 reactions including iso-enzymes, with real stoichiometry $\mathbf{S}$ and measured initial concentrations. Unlike the all- $|s|=1$ toy it is genuinely *multi-substrate* (15 first-order, 13 second-order, one fourth-order), and the rate laws are *Michaelis–Menten* (MM), the saturating law $v = V_{\max} c / (K_m + c)$, so $v_j = V_{\max,j} \prod_k [c_k / (K_{m,kj} + c_k)]^{|S_{kj}|}$ rather than a power law. $V_{\max}$ is the per-reaction rate constant to recover (the MM analogue of k_j) and K_m the half-saturation constant; known $V_{\max}/K_m$ make this the cleanest available ground truth. *Status:* the SBML import, $\mathbf{S}$ extraction, and an MM simulator are in place (12,000-frame dataset), with the connectome-style autoregressive curriculum ported to stabilise the rollout. Two cautions, both found by being sceptical of a good number. A test rollout that looked catastrophic was an evaluator bug (the boundary stimulus was omitted); supplied, the curriculum model reproduces the trajectory well (Fig. 7). But that success is *in-sample*, and both the leak-resistant recovery test (Fig. 8) and a direct held-out rollout (Fig. 7) fail: the model reproduces the dynamics with the *wrong* kinetics, a degenerate fit, and the good in-sample rollout was effectively memorisation of one trajectory. The curriculum fixed rollout *stability*; the open problem on real MM data is parameter *identifiability* — the same scale/compensation degeneracy as the synthetic setup, here unbroken.

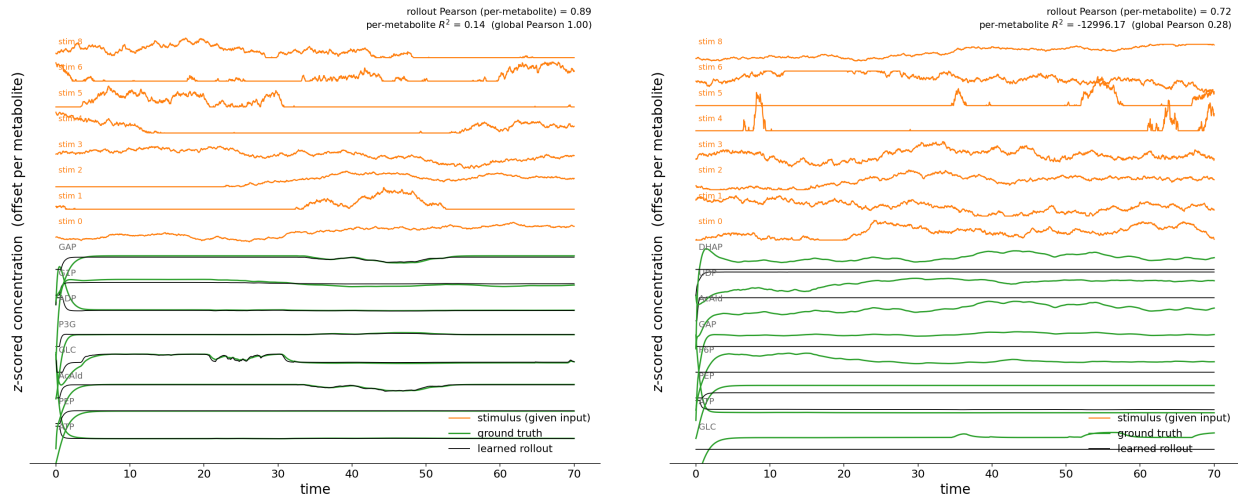


Figure 7: **Rung 1 glycolysis rollout: in-sample vs held-out.** Eight highest-variance metabolites, z -scored and stacked: ground truth (green) vs the learned free autoregressive rollout (dashed black); the given external stimulus channels (orange, top) are the known boundary drive, not predicted. **Left (in-sample):** the rollout tracks the training trajectory. **Right (held-out):** on an unseen trajectory (different stimulus and initial state) the rollout settles to the wrong levels and no longer tracks ground truth.

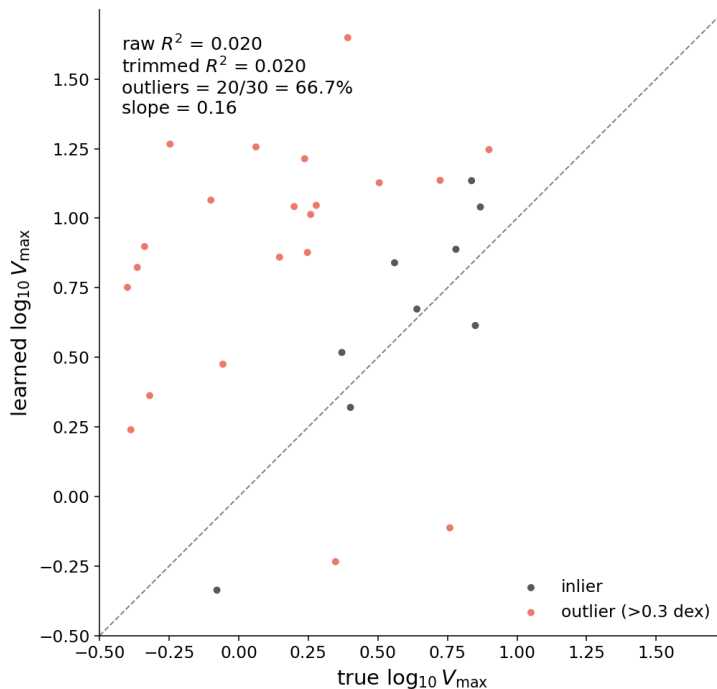


Figure 8: **Rung 1 glycolysis: $V_{\max}$ recovery (leak-resistant test).** Learned vs true $\log_{10} V_{\max}$ for the 30 reactions (dashed: identity); red points exceed the 0.3-dex outlier gate. Despite the near-perfect in-sample rollout (Fig. 7), the recovered kinetics do not track ground truth — a degenerate fit.

Does MLP_{sub} learn the glycolysis kinetics, and does the log-exp form help? A natural suspect for the recovery failure is that MLP_{sub} cannot represent the saturating MM substrate law $[c/(K_m+c)]^{|s|}$. Comparing the learned MLP_{sub} against that truth, for the plain MLP and the log-space form $g = \exp(\text{MLP}(\log c, |s|))$ (Fig. 9), representation is *not* the bottleneck: the well-populated $|s|=1$ and $|s|=2$ shapes are captured, and only the single $|s|=4$ reaction is unconstrained (too few examples to fit). The log-exp form does not rescue recovery either. The limiting factor is not how MLP_{sub} is parameterised but the MM identifiability degeneracy ($V_{\max}$ and K_m trade off): the substrate function can be represented, yet the rate constants are not recovered.

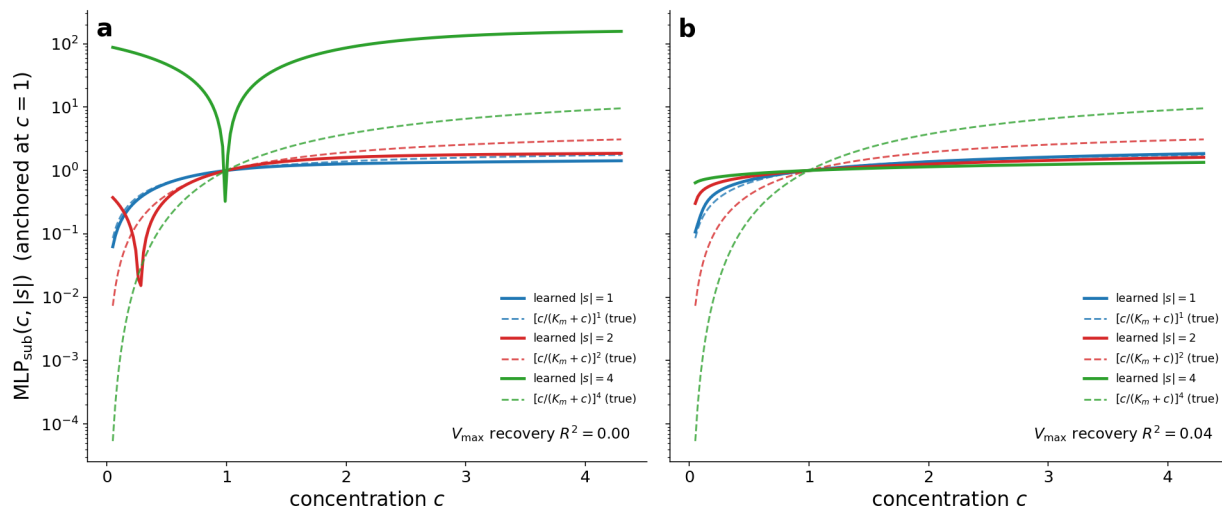


Figure 9: **Rung 1: does MLP_{sub} learn the glycolysis substrate kinetics?** Learned $\text{MLP}_{\text{sub}}(c, |s|)$ (solid) vs the true Michaelis–Menten law $[c/(K_m + c)]^{|s|}$ (dashed) for the substrate orders $|s|=1, 2, 4$, log y -axis, anchored at $c=1$. (a) plain MLP, (b) log-space MLP. The well-populated $|s|=1, 2$ shapes are captured; the single $|s|=4$ reaction is unconstrained.

The limit is signal, not capacity. Two controls on a synthetic $|s| \geq 2$ regime localise the residual failure. Training on four diverse trajectories instead of one barely helps, so it is not a data-volume problem; and a *structured* substrate function $g = c^{a(|s|)}$ — which can represent any exponent exactly — still fails, its learned exponent map nailing $a(1)$ but never reaching the high exponents because those reactions are *rare*. The bottleneck for high-order kinetics is neither MLP capacity nor data volume but the *scarcity of high-order reactions*: too little signal to identify a steep exponent. The fix is a structural prior (e.g. integer exponents) or a regime richer in high- $|s|$ reactions, not a better function approximator.

Rung 2 — consensus genome-scale model (yeast-GEM). The community-curated genome-scale model of *S. cerevisiae* (Lu et al., 2019): 2,806 metabolites, 4,131 reactions, supplying a large real $\mathbf{S}$ and flux bounds but *no* kinetic rate laws. The intended use is to extract a central-carbon subgraph, drive it with dynamic-FBA or literature kinetics, and fit the GNN in “S given” mode on realistic topology. *Status*: not started beyond vendoring; subgraph extraction and a dynamics generator remain to be written.

Rung 3 — real-time metabolomics during nutrient transitions (real data). High-frequency flow-injection mass spectrometry of live microbial culture (Link et al., 2015). For *E. coli* the released data give 247 annotated ions over 119 time points in three replicates, plus a starvation→growth switch series; companion *B. subtilis* and *Y. lipolytica* sets are included. Each ion carries a KEGG annotation, though many are mass-ambiguous and the values are *relative* intensities, not absolute concentrations. *Status*: ingested and characterised (Fig. 10). The trajectory is *information-rich* (activity rank well above the closed synthetic networks), so data richness is *not* the limiting factor. The obstacle is *partial observation*: the measured set is a small, open fraction of iJO1366 whose dynamics depend on unmeasured species; the only fully-observed subnetwork (reactions whose every

metabolite is measured) is ~ 181 of iJO1366’s reactions, the immediate LSQ/GNN target. There is *no* kinetic ground truth, so success can be judged only by held-out prediction. A parallel mammalian direction is chronic jugular microdialysis in freely moving mice (Nardin et al., 2025).

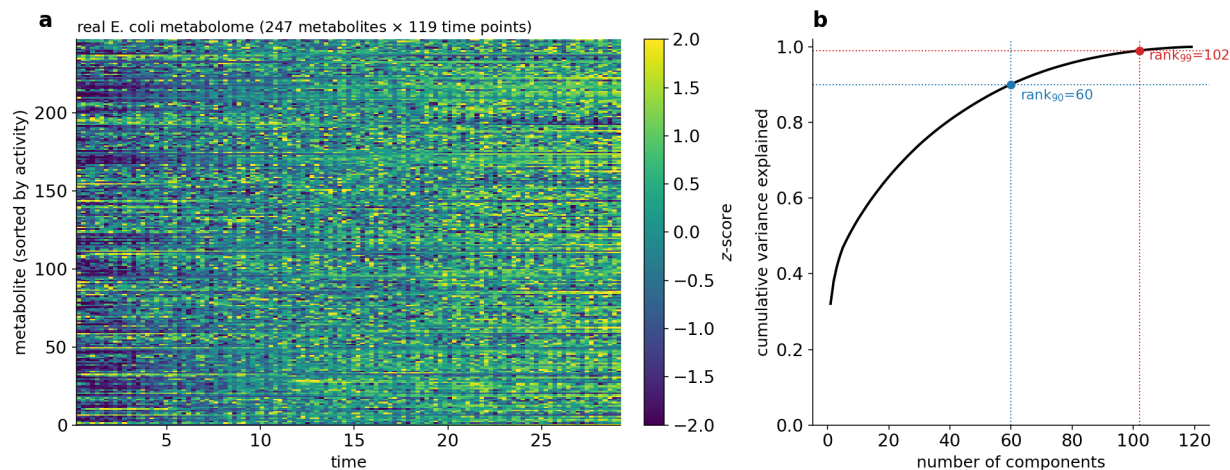


Figure 10: **The real *E. coli* metabolome is a high-rank trajectory.** (a) Measured trajectory: 247 metabolites $\times$ 119 time points (relative ion intensities; z -scored per metabolite, rows sorted by activity). (b) SVD cumulative-variance spectrum (activity rank far richer than the closed synthetic networks).

What transfers and what changes. The core recipe transfers: the bipartite graph, message passing over reactions, function discovery via $\text{MLP}_{\text{sub}}/\text{MLP}_{\text{node}}$, and log-space rate-constant recovery. Real data adds, all at once, the obstacles the synthetic setup omits: saturating/allosteric kinetics that MLP_{sub} must learn; a partial, incomplete $\mathbf{S}$; far fewer, unevenly spaced time points; relative rather than absolute concentrations; and external drivers entering as $f_{\text{ext}}(t)$ in Eq. (2).

Identifiability on a real network topology (imposed kinetics). Short of full real data, we can probe identifiability on *real* topology by imposing synthetic MM kinetics (random V_{max}/K_m , hence known ground truth) on the real $\mathbf{S}$ of two networks: *E. coli* core (72 metabolites, 71 reactions) and a yeast-GEM central-carbon subgraph (208 metabolites, 120 reactions). With $\mathbf{S}$ given the model reproduces the dynamics yet recovers *none* of the kinetics on either network (Fig. 11). The cause is dynamical, not structural: these closed networks relax to a steady state within a few frames, so their trajectories have activity rank ≈ 2 — far below the oscillatory toy. Identifiability tracks the *information content of the dynamics*, not network size or method: a closed subnetwork at steady state is information-starved. Recovering kinetics on real topology requires holding the system *off* its fixed point — a sustained external drive, or the nutrient switch of Rung 3 — to raise the rank.

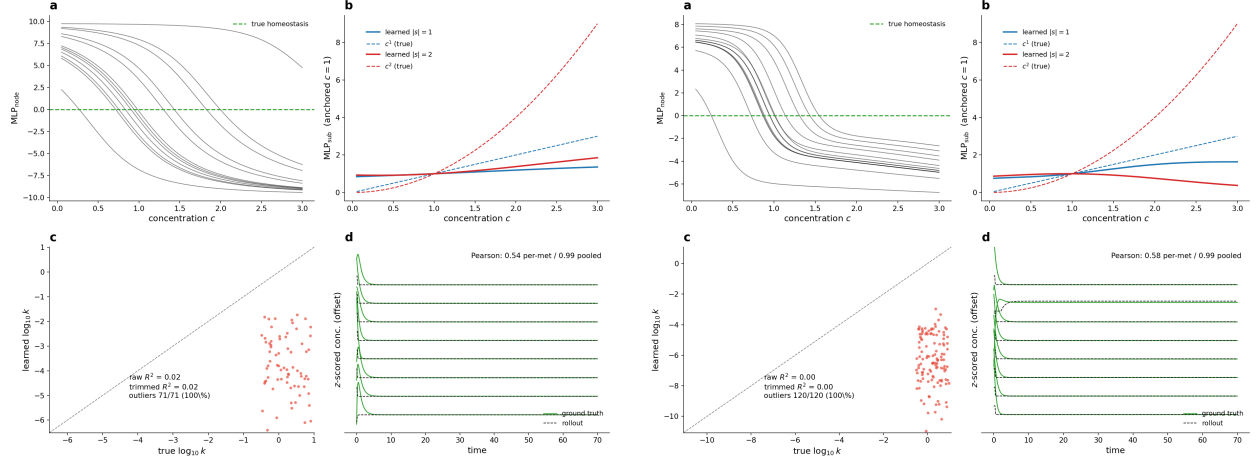


Figure 11: **Identifiability on real topology with imposed MM kinetics.** Left: *E. coli* core (72×71); right: yeast-GEM cytosolic subgraph (208×120). Real stoichiometry, synthetic MM kinetics (so $V_{\max}/K_m$ are known), $\mathbf{S}$ given. Each dashboard: (a) MLP_{node} , (b) MLP_{sub} , (c) $V_{\max}$ recovery, (d) free rollout.

Raising identifiability: drive the network off steady state. The failure above is a *data* problem with a precise linear-algebra reading. With $\mathbf{S}$ given, recovering the kinetics is *linear in the parameters*, $\mathbf{A}\boldsymbol{\theta} = \mathbf{b}$; a metabolite that never varies gives constant or zero columns — null directions in $\mathbf{A}$ — so its parameters are unidentifiable. Equivalently, linearising as $\dot{c} = Jc + Bu(t)$, an autonomous trajectory collapses into a few slow eigenmodes, whereas an external source u on m metabolites drives it into a controllability subspace whose dimension grows with m . Injecting a known multi-frequency metabolite source and sweeping its breadth, the activity rank rises *linearly* with the number of driven metabolites (rank $\approx m$) on both real topologies (Fig. 12), clearing the rank-20 target. Information content, not network size, is the lever; the next step trains on these high-rank, stimulus-driven trajectories.

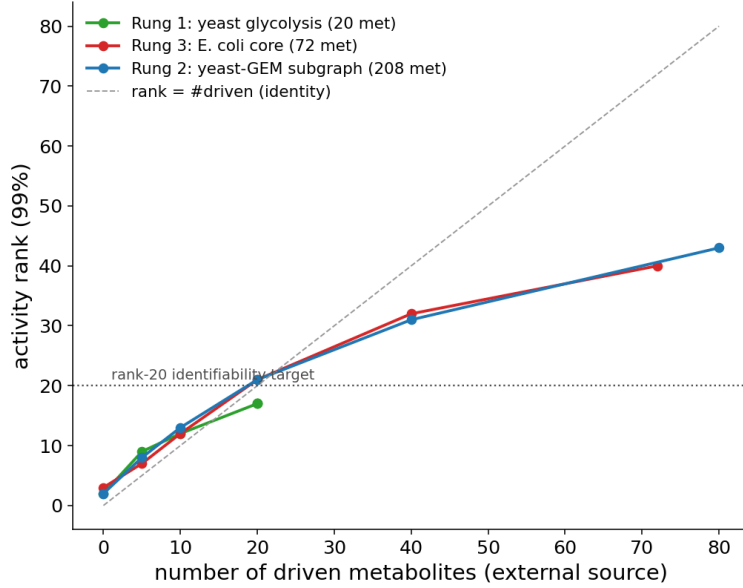


Figure 12: **An external metabolite source raises the activity rank $\approx$ linearly.** Activity rank (99%) of the simulated trajectory vs the number of metabolites carrying a known multi-frequency source drive, for all three rungs under imposed MM kinetics: Rung 1 glycolysis (20 met), Rung 3 *E. coli* core (72 met), Rung 2 yeast-GEM subgraph (208 met). Dashed line: rank $\approx m$ identity.

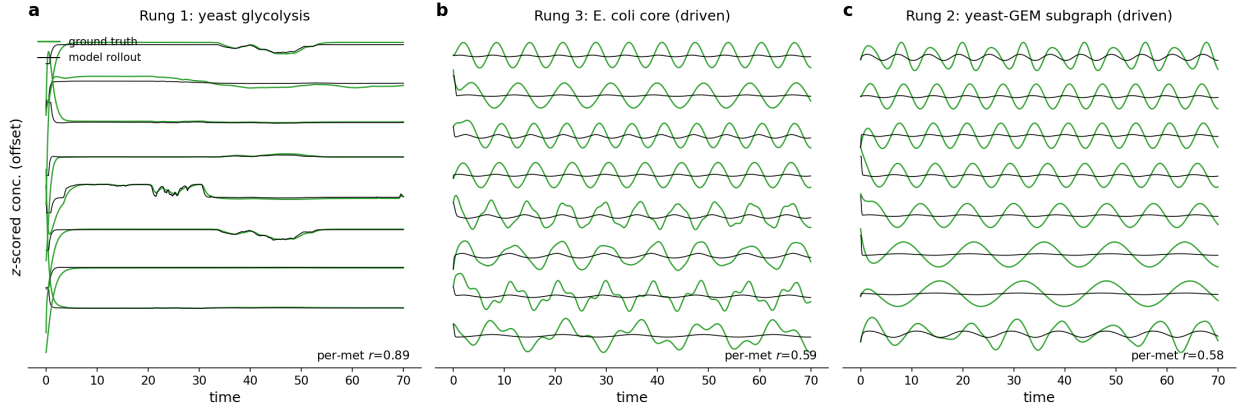


Figure 13: **Free-rollout traces under the external drive, one panel per rung** (the trained GNN). Eight highest-variance metabolites, z -scored and stacked: ground truth (green) vs the model’s free rollout (black). **(a)** Rung 1 glycolysis; **(b)** Rung 3 *E. coli* core and **(c)** Rung 2 yeast-GEM subgraph, both with the $m=40$ metabolite source holding the network in sustained oscillation. Per-metabolite Pearson annotated. The least-squares companion is Fig. 14.

Identifiability is not the bottleneck — learnability is. The GNN trained on the driven (rank-38) *E. coli* data still recovers *none* of the kinetics, yet the inverse problem is *well-posed*: given $\mathbf{S}$ and the rate-law shape, dc/dt is linear in the per-reaction scale, $A\theta = b$ with $A_{:j} = \mathbf{S}_{:j} \phi_j(c)$, and built from the simulator’s Jacobian this A is full column rank and well conditioned at both rank-2 and rank-38 — so a plain least-squares solve recovers $V_{\max}$ exactly and rolls it out perfectly (Fig. 14).

The catch is that this is conditional on *knowing* ϕ ; the GNN must learn the shape (MLP_{sub}) and the scale (V_{max}) jointly, and a saturating shape with a free K_m can absorb the scale so V_{max} drifts. GNN success is thus set by shape $\times$ scale learnability, not identifiability. The lever is the model, not the data: solve V_{max} by least squares given the learned shape, or constrain MLP_{sub} — not more rank, drive, or trajectories.

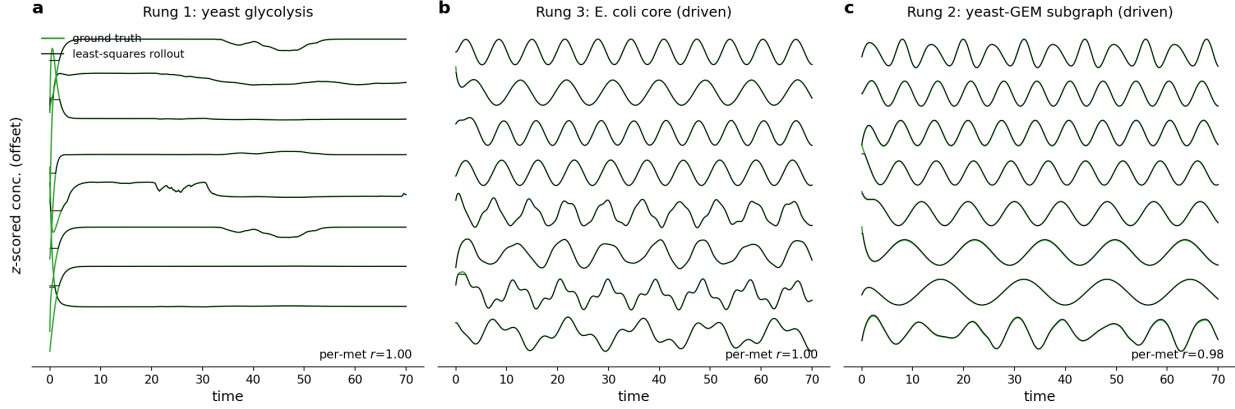


Figure 14: **Least-squares rollout (companion to Fig. 13).** Free-rollout traces for the same three rungs and external drive as Fig. 13, but with the per-reaction V_{max} set by a least-squares solve given the true substrate shape rather than by the GNN. Eight highest-variance metabolites, z -scored and stacked: ground truth (green) vs the least-squares rollout (black). **(a)** Rung 1 glycolysis; **(b)** Rung 3 *E. coli* core; **(c)** Rung 2 yeast-GEM subgraph. Per-metabolite Pearson annotated.

A realistic, non-oscillatory stimulus changes nothing. One might worry the failure is an artefact of the *shape* of the drive — a clean multi-sinusoid is nothing like a real metabolome. Replacing the sinusoid with an Ornstein–Uhlenbeck / AR(1) drive ($\phi=0.98$), smooth and aperiodic like the measured *E. coli* trajectory, and regenerating all three rungs with the same imposed-MM kinetics reproduces the failure *exactly*: the GNN still recovers V_{max} at $R^2 \approx 0$ while rolling out serviceably. Least squares, by contrast, recovers and free-rolls-out the dynamics almost perfectly on the same realistic-stimulus data (Fig. 15), exactly as under the sinusoid (Fig. 14). The learnability barrier is *stimulus-shape-invariant*: neither the richness nor the temporal character of the drive is the lever. The lever remains model-side.

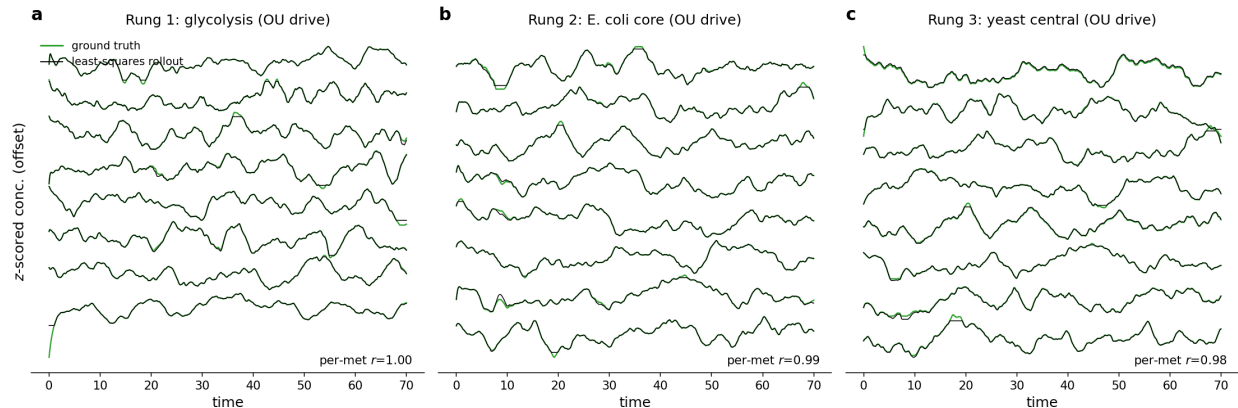


Figure 15: **Least-squares rollout under a realistic (OU) stimulus** (companion to Fig. 14). As Fig. 14, but on the realistic Ornstein–Uhlenbeck / AR(1)-driven rungs: ground truth (green) vs the least-squares rollout (black), z-scored and stacked. **(a)** Rung 1 glycolysis; **(b)** Rung 2 *E. coli* core; **(c)** Rung 3 yeast central. Per-metabolite Pearson annotated.

Least squares does the job; the GNN hardly does it yet. Putting the two estimators side by side on the same realistic-stimulus data (Figs. 16 and 17) locates the difficulty. Given the true substrate shape, least squares recovers $V_{\max}$ exactly on the two rungs where the dynamics are linear in $V_{\max}$ (glycolysis and *E. coli* core); the third (yeast central) is flux-limit-clamped, a simulator artefact, not a data limit. The GNN, which must *discover* shape and scale jointly, recovers essentially none of it. The remaining gap is closed model-side: feed the GNN the shape (or constrain MLP_{sub}) and solve the scale by least squares, rather than asking it to learn both at once.

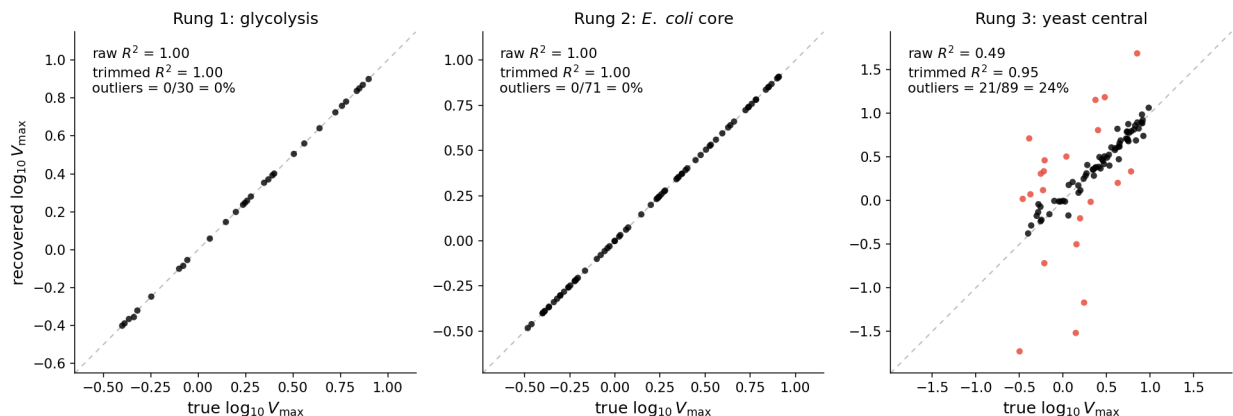


Figure 16: **Per-reaction $V_{\max}$ recovery — least squares.** Recovered vs true $\log_{10} V_{\max}$ for every reaction (least squares given the true substrate shape), one panel per rung: **(a)** glycolysis, **(b)** *E. coli* core, **(c)** yeast central. Dashed line: exact recovery; black inliers, red points exceed the 0.3-dex outlier gate (raw and trimmed R^2 and outlier fraction annotated). Companion to Fig. 17.

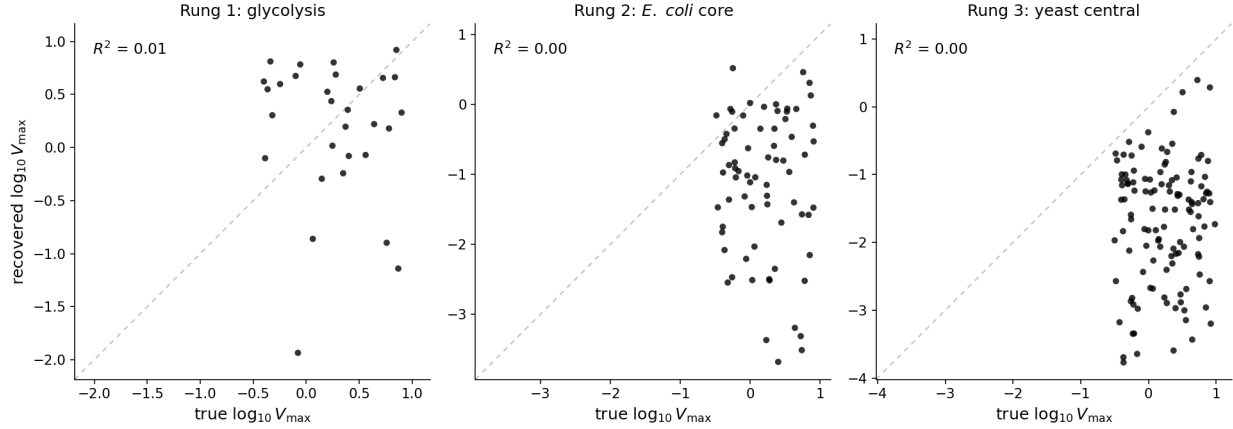


Figure 17: **Per-reaction $V_{\max}$ recovery — the trained GNN.** As Fig. 16, for the GNN’s leak-resistant recovery: recovered vs true $\log_{10} V_{\max}$ per reaction, one panel per rung. Dashed line: exact recovery.

Target	Network / kinetics	Ground truth	Status
Synthetic k_j (S given)	random / mass-action	k_j known	works , $R^2 \approx 0.87$
Synthetic homeostasis	random / mass-action	λ_i known	open (single-step blind)
Rung 1: yeast glycolysis	real / MM+Hill	$V_{\max}, K_m$ known	rollout in-sample; $V_{\max}$ $R^2 \approx 0$ (degenerate)
Real topology (<i>E. coli</i> core)	real S / imposed MM	$V_{\max}$ known	$V_{\max}$ $R^2 \approx 0$, 100% out (rank-2 degenerate)
Real topology (yeast-GEM subgraph)	real S / imposed MM	$V_{\max}$ known	$V_{\max}$ $R^2 \approx 0$, 100% out (rank-2 degenerate)
+ realistic OU drive (rungs 1–3)	real S / imposed MM	$V_{\max}$ known	$V_{\max}$ $R^2 \approx 0$; rollout 0.83–0.90 (shape-invariant)
Rung 3: real <i>E. coli</i> data	inferred / unknown	none	ingested; rank 102, 181 fully-obs. rxns; LSQ/GNN next

Table 4: Where each target stands. Only the first row is a validated, leak-free result; the rest is the work ahead.

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